



where R is 120Ω termination resistor.

### 3、CAN communication data format

The CAN network signal data format follows the following table, and the two-line definition follows the first line;

| data type                            | Scale factor | Range (actual range).   | The offset | The number of bytes |
|--------------------------------------|--------------|-------------------------|------------|---------------------|
| Data format definitions for each ECU |              |                         |            |                     |
| Total voltage (DC)                   | 0.1V/bit     | 0—5000(0—500V)          | 0          | 2BYTE               |
| Total current (DC)                   | 0.1A/bit     | 0—8000( -400A) — 400A ) | -400A      | 2BYTE               |
| Phase current                        | 1A/bit       | 1300—4100(-900A-900A )  | -3200A     | 2BYTE               |
| Monomer voltage                      | 0.001V/bit   | 0—6000(0—6V)            | 0          | 2BYTE               |
| PWM pulse width                      | 1Hz/bit      | 0-65500(0-65500Hz)      | 0          | 2BYTE               |
| Insulation resistance value          | 1kΩ/bit      | 0-30000(0-30M)          | 0          | 2BYTE               |
| temperature                          | 1℃/bit       | 0—190 ( -40—150℃ )      | -40        | 1BYTE               |
| Battery (SOC).                       | 1%/bit       | 0—100 ( 0—100% )        | 0          | 1BYTE               |

### 4、Communication specifications

#### 4.1 Principles to be followed at the data link layer

The bus communication rate is: 250Kbps. The data link layer specification mainly refers to the relevant provisions of CAN2.0B and SAE J1939. The 29-bit identifier of the frame is extended using CAN and redefined, and the following is the allocation table for the 29-bit identifier:

| IDENTIFIER 11BITS |    |    |    |    |                |    |    |    |    |    | SRR | Goes |
|-------------------|----|----|----|----|----------------|----|----|----|----|----|-----|------|
| PRIORITY          |    |    | R  | DP | PDU FORMAT(PF) |    |    |    |    |    | SRR | Goes |
| 3                 | 2  | 1  | 1  | 1  | 8              | 7  | 6  | 5  | 4  | 3  |     |      |
| 28                | 27 | 26 | 25 | 24 | 23             | 22 | 21 | 20 | 19 | 18 |     |      |

| IDENTIFIER EXTENSION 18BITS |    |                  |    |    |    |    |    |   |   |                    |   |   |   |   |   |   |   |
|-----------------------------|----|------------------|----|----|----|----|----|---|---|--------------------|---|---|---|---|---|---|---|
| PF                          |    | PDU SPECIFIC(PS) |    |    |    |    |    |   |   | SOURCE ADDRESS(SA) |   |   |   |   |   |   |   |
| 2                           | 1  | 8                | 7  | 6  | 5  | 4  | 3  | 2 | 1 | 8                  | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| 17                          | 16 | 15               | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

where the priority is 3 bits, there can be 8 priorities, R is generally fixed to 0, DP is now fixed to 0, 8 bits PF is the code of the message, 8-bit PS is the destination address or group extension, and 8-bit SA is the source address where this message is sent.

›Each node connected to the network has a PGN and address, PGN is used to identify the function of the node and conduct address arbitration, and the address is used for data communication of the node.

›Each node has at least one function, there may be multiple nodes with the same function, or a node with

multiple functions

›For multibyte data, use a little-endian approach, such as 4660=0x1234, first send 0x 34, then 0x12

## 4.2 CAN Network Address Allocation Table

| The node name address     | Address SOURCE ADDRESS(SA) |
|---------------------------|----------------------------|
| Battery management system | 244 ( 0xF4 )               |
| Motor controller          | 240 ( 0XF0 )               |
| Gauge display controller  | 166 ( 0XA6 )               |
| Multimedia controller     | 251 ( 0XFB )               |
| Car charger               | 229 ( 0xE5 )               |
| broadcast address         | 80(0x50)                   |
| Vehicle controller        | 20(0x14)                   |

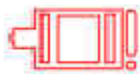
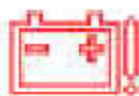
## 4.3 Vehicle network communication hardware requirements

- 1), the network system of the 3 subnets CAN1, CAN2, INCAN is physically completely isolated;
- 2) The network system supports hot swapping. The power supply should have reverse connection protection and power failure detection function;
- 3), the communication cable of the CAN bus adopts shielded twisted pair (flame retardant 0.5nun), the shielding layer should be connected to the CAN\_GND, and the grounding method of the shielded wire is selected by the vehicle when wiring;
- 4) Each component on the CAN bus has a terminal resistance (120Ω ). At the same time, the terminal resistor and the network cable are connected by jumpers for flexible collocation, easy to debug and use, and removed when loading;
- 5), the terminal resistance head (120Ω) is installed at both ends of the network cable and is made in the wire bundle;
- 6), CAN communication cable should leave the power line (more than 0.5m), leave the 12V control line (0.1m or more);
- 7), the cable shield layer is continuously turned on in the car, it is recommended that the network socket of each component has a shielded connector.

## 5、 Protocol messages

### 5.1 MCU sending

#### 5.1.1 MCU sends instrument message 1

| ID          | P   | PGN                                         |                                                                                                                                                                                                                                                                                                                                                                  |            |                     | His | Period (ms). |
|-------------|-----|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------|-----|--------------|
| 0x 1801A6F0 | 6   | R                                           | DP                                                                                                                                                                                                                                                                                                                                                               | PF         | PS                  | 240 | 50ms         |
|             |     | 0                                           | 0                                                                                                                                                                                                                                                                                                                                                                | 01         | 166                 |     |              |
| Data        |     |                                             |                                                                                                                                                                                                                                                                                                                                                                  |            |                     |     |              |
| location    |     | The data name                               | resolution                                                                                                                                                                                                                                                                                                                                                       | The offset | The range of values |     |              |
| BYTE        | BIT |                                             |                                                                                                                                                                                                                                                                                                                                                                  |            |                     |     |              |
| BYTE0       | 1-0 | The motor controller is in a state of state | Bit1 Bit0<br><br>0 0: Normal ready to run state (instrument lit up  );<br>0 1: Motor controller system failure; [For specific faults, see BYTE2-BYTE4, the motor controller does not output power.]<br>Where the fault in BYTE2, the controller cuts off the main contactor]. |            |                     |     |              |
| BYTE0       | 7-2 | retain                                      | All locations 1                                                                                                                                                                                                                                                                                                                                                  |            |                     |     |              |
| BYTE1       | 1-0 | Forced downtime failure                     | Bit1 Bit0<br><br>0 0: No forced shutdown;<br>0 1: Forced shutdown<br>In the following cases, the controller needs to gradually turn off the power within 30S and output N gears:<br>1. Loss of communication contact with BMS 30S;<br>2. Power battery related three-level hugging (see BMS part).                                                               |            |                     |     |              |
| BYTE1       | 3-2 | Controller hardware failure                 | Bit3 Bit2<br><br>0 0: Normal;<br><br><br>0 1: Fault (meter illuminated  ).                                                                                                                                                                                                  |            |                     |     |              |
| BYTE1       | 5-4 | Motor controller overcurrent alarm          | Bit5 Bit4<br><br>0 0: Normal;<br><br><br>0 1: Fault (meter lighting  ) overcurrent [400A].<br>0x2: Hall signal failure (instrument point  ).                                             |            |                     |     |              |
| BYTE1       | 7-6 | Busbar voltage status                       | Bit7 Bit6<br><br>0 0: Normal;<br><br><br>0 1: Busbar undervoltage (meter lighting  ).<br><br><br>1 0: Busbar overvoltage (meter lighting  ).                                           |            |                     |     |              |

|       |     |                                                  |                                                                                                                                                                                                                                                                                                  |   |   |
|-------|-----|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| BYTE2 | 1-0 | Gear failure                                     | Bit1 Bit0<br>0 0: Normal;<br>0 1: Fault (meter illuminated  ).                                                                                                                                                |   |   |
| BYTE2 | 3-2 | Communication failure                            | Bit3 Bit2<br>0 0: Normal;<br>0 1: Fault (meter illuminated  ).                                                                                                                                                |   |   |
| BYTE2 | 5-4 | Accelerator pedal failure                        | Bit5 Bit4<br>0 0: Normal;<br>0 1: Fault (meter illuminated  ).                                                                                                                                                |   |   |
| BYTE2 | 7-6 | The fault level of the electronic control system | Bit7 Bit6<br>0 0: no faults;<br>0 1: first-level failure;<br>1 0: Secondary failure;<br>1 1: three-level failure;                                                                                                                                                                                |   |   |
| BYTE3 | 1-0 | Overtemperature alarm                            | Bit1 Bit0<br>0 0: Normal;<br>0 1: Controller overheating (meter point  );<br>1 0: The motor overheats (meter point  ). |   |   |
| BYTE3 | 3-2 | Overload alarm                                   | Bit3 Bit2<br>0 0: Normal;<br>0 1: Controller overload (instrument point  );<br>1 0: Motor overload (meter point  ).    |   |   |
| BYTE3 | 5-4 | Number of drive motors                           | 1                                                                                                                                                                                                                                                                                                | 0 | 3 |
| BYTE3 | 7-6 | Drive motor serial number                        | 1                                                                                                                                                                                                                                                                                                | 0 | 3 |
| BYTE4 | 1-0 | Gear signal                                      | Bit1 Bit0<br>0 0: N gear;<br>0 1: R gear;<br>1 0: D gear;<br>1 1: S gear;<br>(When the gear is faulty, the controller outputs a N signal.)<br>P file reserved).                                                                                                                                  |   |   |

|       |     |                                |                                                                                                                                    |   |        |
|-------|-----|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---|--------|
| BYTE4 | 2   | Braking system failure         | 0: Normal<br>1: Braking system failure                                                                                             |   |        |
| BYTE4 | 4-3 | Actual steering                | Bit4 Bit3<br>0 1: CCW direction (counterclockwise);<br>1 0: CW direction (clockwise);                                              |   |        |
| BYTE4 | 7-5 | Actual operating status        | Bit7 Bit6 Bit5<br>0 0 0: Stop<br>0 0 1: Power consumption<br>0 1 0: Power generation<br>0 1 1: Ready<br>1 0 0: Exception (faulty). |   |        |
| BYTE5 | 7-0 | Total number of motor failures | 1/bit                                                                                                                              | 0 | 0-255  |
| BYTE6 | 0   | Brake pedal signal             | 0: No brakes<br>1: There are brakes                                                                                                |   |        |
| BYTE6 | 7-1 | Accelerator pedal opening      | 1%                                                                                                                                 | 0 | 0-100% |
| BYTE7 | 7-0 | List of motor fault codes      | 1                                                                                                                                  | 0 | 0-255  |

### 5.1.2 MCU sends instrument message 2

| ID          | P   | PGN                                          |            |            |                                                | His | Period (ms). |
|-------------|-----|----------------------------------------------|------------|------------|------------------------------------------------|-----|--------------|
| 0x 1802A6F0 | 6   | R                                            | DP         | PF         | PS                                             | 240 | 50ms         |
|             |     | 0                                            | 0          | 02         | 166                                            |     |              |
| Data        |     |                                              |            |            |                                                |     |              |
| location    |     | The data name                                | resolution | The offset | The range of values                            |     |              |
| BYTE        | BIT |                                              |            |            |                                                |     |              |
| BYTE0       | 7-0 | Busbar voltage low bytes                     | 0.1V       | 0          | 0-500V                                         |     |              |
| BYTE1       | 7-0 | Busbar voltage high bytes                    |            |            |                                                |     |              |
| BYTE2       | 7-0 | Busbar current low bytes                     | 0.1A       | -320       | -320-400A [negative value indicates charging]. |     |              |
| BYTE3       | 7-0 | Busbar current high bytes                    |            |            |                                                |     |              |
| BYTE4       | 7-0 | The actual torque of the motor is low bytes  | 0.1N· m    | 0          | 0-300N· m                                      |     |              |
| BYTE5       | 7-0 | The actual torque of the motor is high bytes |            |            |                                                |     |              |
| BYTE6       | 7-0 | The actual speed of the motor is low         | 1.0rpm     | 0          | 0-11000rpm                                     |     |              |
| BYTE7       | 7-0 | The actual                                   |            |            |                                                |     |              |

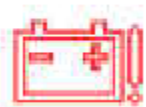
|  |  |                            |  |  |  |
|--|--|----------------------------|--|--|--|
|  |  | speed of the motor is high |  |  |  |
|--|--|----------------------------|--|--|--|

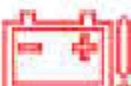
### 5.1.2 MCU sends instrument message 3

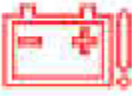
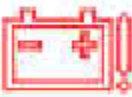
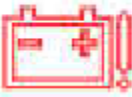
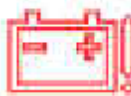
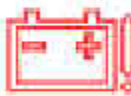
| ID          | P   | PGN                                   |            |            |                                     | His | Period (ms). |
|-------------|-----|---------------------------------------|------------|------------|-------------------------------------|-----|--------------|
| 0x 1803A6F0 | 6   | R                                     | DP         | PF         | PS                                  | 240 | 50ms         |
|             |     | 0                                     | 0          | 03         | 166                                 |     |              |
| Data        |     |                                       |            |            |                                     |     |              |
| location    |     | The data name                         | resolution | The offset | The range of values                 |     |              |
| BYTE        | BIT |                                       |            |            |                                     |     |              |
| BYTE0       | 7-0 | The motor phase voltage is low bytes  | 0.1V       | 0          | 0-500V                              |     |              |
| BYTE1       | 7-0 | The motor phase voltage is high       |            |            |                                     |     |              |
| BYTE2       | 7-0 | The motor phase current is low bytes  | 0.1A       | -3200      | -900-900A                           |     |              |
| BYTE3       | 7-0 | The motor phase current is high bytes |            |            |                                     |     |              |
| BYTE4       | 7-0 | Low speed bytes                       | 0.15km/h   | 0          | 0km/h-150km/h                       |     |              |
| BYTE5       | 7-0 | Vehicle speed high bytes              |            |            |                                     |     |              |
| BYTE6       | 7-0 | Controller temperature                | 1℃         | -40        | -40-150℃                            |     |              |
| BYTE7       | 7-0 | Motor temperature                     | 1℃         | -40        | -40-150℃ (no motor temperature FF). |     |              |

### 5.2 BMS sending

#### 5.2.1 BMS sends instrument message 1

| ID          | P   | PGN                      |                                                                                                                                                                                                                                                                                                                 |            |                     | His | Period (ms). |
|-------------|-----|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------|-----|--------------|
| 0x 1801A6F0 | 6   | R                        | DP                                                                                                                                                                                                                                                                                                              | PF         | PS                  | 240 | 50ms         |
|             |     | 0                        | 0                                                                                                                                                                                                                                                                                                               | 01         | 166                 |     |              |
| Data        |     |                          |                                                                                                                                                                                                                                                                                                                 |            |                     |     |              |
| location    |     | The data name            | resolution                                                                                                                                                                                                                                                                                                      | The offset | The range of values |     |              |
| BYTE        | BIT |                          |                                                                                                                                                                                                                                                                                                                 |            |                     |     |              |
| BYTE0       | 1-0 | Single overvoltage alarm | <div>Bit1 Bit0</div> <div>0 0: Normal</div> <div>0 1: Level 1 alarm</div> <div></div> <div>1 0: Secondary alarm (instrument lighting  )</div> <div>[in accordance with the requirements of the technical agreement].</div> |            |                     |     |              |

|       |     |                           |                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BYTE0 | 3-2 | Single undervoltage alarm | <p>Bit3 Bit2<br/>0 0: Normal</p> <p>0 1: Level 1 alarm (meter lit  ).</p> <p>1 0: Secondary alarm (instrument lighting  ) [in accordance with the requirements of the technical agreement].</p>     |
| BYTE0 | 5-4 | Insulation alarm          | <p>Bit5 Bit4<br/>0 0: Normal</p> <p>0 1: Level 1 alarm (meter lit  ).</p> <p>1 0: Secondary alarm (instrument lighting  ) [in accordance with the requirements of the technical agreement].</p>     |
| BYTE0 | 7-6 | Battery temperature       | <p>Bit7 Bit6<br/>0 0: Normal</p> <p>0 1: Level 1 alarm (meter lit  ).</p> <p>1 0: Secondary alarm (instrument lighting  ) [in accordance with the requirements of the technical agreement].</p> |
| BYTE1 | 1-0 | Main relay status         | <p>Bit1 Bit0</p> <p>0 0: Disconnect (meter lit  ).</p> <p>0 1: Close</p>                                                                                                                                                                                                             |
| BYTE1 | 3-2 | retain                    | The default is sent 11                                                                                                                                                                                                                                                                                                                                                    |
| BYTE1 | 5-4 | retain                    | The default is sent 11                                                                                                                                                                                                                                                                                                                                                    |
| BYTE1 | 6   | retain                    | The default is sent 1                                                                                                                                                                                                                                                                                                                                                     |
| BYTE1 | 7   | SOC low alarm             | <p>Bit7<br/>0: Normal</p> <p>1: SOC low alarm (meter lit  ).</p> <p>【In accordance with the requirements of the technical agreement】</p>                                                                                                                                             |
| BYTE2 | 1-0 | retain                    | The default is sent 11                                                                                                                                                                                                                                                                                                                                                    |

|       |     |                                              |                                                                                                                                                                                                                                                                                                                                                            |   |                              |
|-------|-----|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|
| BYTE2 | 3-2 | retain                                       | The default is sent 11                                                                                                                                                                                                                                                                                                                                     |   |                              |
| BYTE2 | 5-4 | Overcurrent (including short circuit).       | Bit5 Bit4<br>0 0: Normal<br><br>0 1: Alarm (meter lit ).<br><b>【Instrument alarm light flashes twice per second and alarms】</b>                                                                                                                                          |   |                              |
| BYTE2 | 7-6 | The battery pack voltage is uneven           | Bit7 Bit6<br>0 0: Normal<br><br>0 1: Alarm (meter lit ).<br><b>【In accordance with the requirements of the technical agreement】</b>                                                                                                                                      |   |                              |
| BYTE3 | 1-0 | Communication failure                        | Bit1 Bit0<br>0 0: Normal<br><br>0 1: Alarm (meter lit ).<br><b>【Communication failure with the vehicle bus】</b>                                                                                                                                                         |   |                              |
| BYTE3 | 3-2 | BMS hardware failure                         | Bit3 Bit2<br>0 0: Normal<br><br>0 1: Alarm (meter lit ).                                                                                                                                                                                                               |   |                              |
| BYTE3 | 5-4 | Total battery pack voltage abnormality alarm | Bit3 Bit2<br>0 0: Normal<br><br>0 1: Undervoltage (meter lit ).<br><br>1 0: Overvoltage (meter lit ).<br><b>【In accordance with the requirements of the technical agreement】</b> |   |                              |
| BYTE3 | 7-6 | Failure level                                | 1                                                                                                                                                                                                                                                                                                                                                          | 0 | 0-3<br>0 indicates no faults |
| BYTE4 | 0   | retain                                       | The default is sent 11                                                                                                                                                                                                                                                                                                                                     |   |                              |
| BYTE4 | 1   | The number of chargeable energy storage      | 1                                                                                                                                                                                                                                                                                                                                                          | 0 | Long 1                       |

|       |     |                                                    |                                                                                                                                                                                                                                                                                                                                                                  |   |                                                                                                                                                                                 |
|-------|-----|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     | subsystems                                         |                                                                                                                                                                                                                                                                                                                                                                  |   |                                                                                                                                                                                 |
| BYTE4 | 2   | Rechargeable energy storage system mismatch alarm  | Bit2<br>0: Normal<br>1: Alarm (matches the battery code stored inside the motor controller, if it is inconsistent, the alarm).                                                                                                                                                                                                                                   |   |                                                                                                                                                                                 |
| BYTE4 | 3   | The number of chargeable energy storage subsystems | 1                                                                                                                                                                                                                                                                                                                                                                | 0 | Long 1                                                                                                                                                                          |
| BYTE4 | 5-4 | The highest failure level of the whole vehicle     | 1                                                                                                                                                                                                                                                                                                                                                                | 0 | Compare battery and electronic control system failure levels and report large numbers<br>Battery:<br>0x1801A6F0/BYTE3/7-6<br>Electronically controlled:<br>0x1801A6F0/BYTE2/7-6 |
| BYTE4 | 7-6 | Temperature difference alarm                       | Bit7 Bit6<br>0 0: Normal<br><br>0 1: Level 1 alarm (meter lit  ).<br><br>1 0: Secondary alarm (instrument lighting  ) [in accordance with the requirements of the technical agreement]. |   |                                                                                                                                                                                 |
| BYTE5 | 0   | retain                                             | The default is sent 11                                                                                                                                                                                                                                                                                                                                           |   |                                                                                                                                                                                 |
| BYTE5 | 2-1 | Run mode                                           | Bit3 Bit2<br>0 0: Pure electric (long-positioned).<br>0 1: Mixing<br>1 0: Fuel<br>1 1: Exception                                                                                                                                                                                                                                                                 |   |                                                                                                                                                                                 |
| BYTE5 | 5-3 | Charging status                                    | Bit5 Bit4 Bit3<br>0 0 0: Parking charging (fast charge or slow charge).<br>0 0 1: Driving charging<br>0 1 0: Uncharged state (not charged).<br>0 1 1: Charging completed (sent when charging is complete).<br>1 0 0: Exception (an exception occurred during charging).                                                                                          |   |                                                                                                                                                                                 |
| BYTE5 | 7-6 | Vehicle status                                     | Bit7 Bit6<br>0 0: Start status (high voltage on, and battery, electronic control are no alarm).<br>0 1: Turn off the ignition (high voltage power-down).<br>1 0: Other states<br>1 1: Abnormal (faulty power battery or motor controller).                                                                                                                       |   |                                                                                                                                                                                 |

|       |     |                                  |   |   |                                  |
|-------|-----|----------------------------------|---|---|----------------------------------|
| BYTE6 | 7-0 | Total number of battery failures | 1 | 0 | 0-255                            |
| BYTE7 | 7-0 | List of battery failure codes    | 1 | 0 | 0-255 (cyclic send, no fault 0). |

### 5.2.2 BMS sends instrument messages 2

| ID          | P   | PGN                                 |            |            |                                                                          | His | Period (ms). |
|-------------|-----|-------------------------------------|------------|------------|--------------------------------------------------------------------------|-----|--------------|
| 0x 1802A6F0 | 6   | R                                   | DP         | PF         | PS                                                                       | 240 | 100ms        |
|             |     | 0                                   | 0          | 02         | 166                                                                      |     |              |
| Data        |     |                                     |            |            |                                                                          |     |              |
| location    |     | The data name                       | resolution | The offset | The range of values                                                      |     |              |
| BYTE        | BIT |                                     |            |            |                                                                          |     |              |
| BYTE0       | 7-0 | Total voltage low bytes             | 0.1V       | 0          | 0-500V (the meter shows this voltage).                                   |     |              |
| BYTE1       | 7-0 | Total voltage high bytes            |            |            |                                                                          |     |              |
| BYTE2       | 7-0 | Total current is low bytes          | 0.1A       | -320       | -320-400A (negative value indicates charging, meter shows this current). |     |              |
| BYTE3       | 7-0 | Total current high bytes            |            |            |                                                                          |     |              |
| BYTE4       | 7-0 | Insulation resistance is low bytes  | 1kΩ        | 0          | 0-30MΩ (insulation detection module device).                             |     |              |
| BYTE5       | 7-0 | Insulation resistance is high bytes |            |            |                                                                          |     |              |
| BYTE6       | 7-0 | SOC value                           | 1%         | 0          | 0-100                                                                    |     |              |
| BYTE7       | 7-0 | Number of battery strings           | 1          | 0          | 0-255                                                                    |     |              |

### 5.2.3 BMS sends instrument messages 3

| ID          | P   | PGN                                                |                |            |                     | His | Period (ms). |
|-------------|-----|----------------------------------------------------|----------------|------------|---------------------|-----|--------------|
| 0x 1803A6F0 | 6   | R                                                  | DP             | PF         | PS                  | 240 | 100ms        |
|             |     | 0                                                  | 0              | 03         | 166                 |     |              |
| Data        |     |                                                    |                |            |                     |     |              |
| location    |     | The data name                                      | resoluti<br>on | The offset | The range of values |     |              |
| BYTE        | BIT |                                                    |                |            |                     |     |              |
| BYTE0       | 7-0 | Single maximum battery voltage low bytes           | EV0.001        | 0          | 0-6V                |     |              |
| BYTE1       | 7-0 | The highest battery voltage of the monomer is high |                |            |                     |     |              |
| BYTE2       | 7-0 | Highest single battery number                      | 1              | 0          | 0-255               |     |              |
| BYTE3       | 7-0 | Monomer minimum battery                            | EV0.001        | 0          | 0-6V                |     |              |

|       |     |                                              |   |   |               |
|-------|-----|----------------------------------------------|---|---|---------------|
|       |     | voltage low bytes                            |   |   |               |
| BYTE4 | 7-0 | Single minimum battery voltage high bytes    |   |   |               |
| BYTE5 | 7-0 | Minimum single battery number                | 1 | 0 | 0-255         |
| BYTE6 | 7-0 | SOH value                                    | 1 | 0 | 0-255         |
| BYTE7 | 1-0 | The minimum temperature subsystem number     | 1 | 0 | 0-3 (long 1). |
| BYTE7 | 3-2 | Maximum temperature subsystem number         | 1 | 0 | 0-3 (long 1). |
| BYTE7 | 5-4 | Lowest voltage battery subsystem number      | 1 | 0 | 0-3 (long 1). |
| BYTE7 | 7-6 | The highest voltage battery subsystem number | 1 | 0 | 0-3 (long 1). |

#### 5.2.4 BMS sends instrument messages 4

| ID          | P   | PGN                                                 |                                                                     |            |                     | His | Period (ms). |
|-------------|-----|-----------------------------------------------------|---------------------------------------------------------------------|------------|---------------------|-----|--------------|
| 0x 1804A6F0 | 6   | R                                                   | DP                                                                  | PF         | PS                  | 240 | 100ms        |
|             |     | 0                                                   | 0                                                                   | 04         | 166                 |     |              |
| Data        |     |                                                     |                                                                     |            |                     |     |              |
| location    |     | The data name                                       | resolution                                                          | The offset | The range of values |     |              |
| BYTE        | BIT |                                                     |                                                                     |            |                     |     |              |
| BYTE0       | 7-0 | Maximum battery temperature                         | 1℃                                                                  | -40        | -40-150℃            |     |              |
| BYTE1       | 7-0 | Battery maximum temperature sensing number          | 1                                                                   | 0          | 0-255               |     |              |
| BYTE2       | 7-0 | Minimum battery temperature                         | 1℃                                                                  | -40        | -40-150℃            |     |              |
| BYTE3       | 7-0 | Battery minimum temperature sensing number          | 1                                                                   | 0          | 0-255               |     |              |
| BYTE4       | 7-0 | The maximum discharge current allowed is low bytes  | 0.1A                                                                | 0          | 0-400A              |     |              |
| BYTE5       | 7-0 | The maximum discharge current allowed is high bytes |                                                                     |            |                     |     |              |
| BYTE6       | 3-0 | Cooling method                                      | Bit3 Bit2 Bit1 Bit0<br>0 0 0 0: Free cooling<br>0 0 0 1: Air cooled |            |                     |     |              |

|       |     |                              |                                                                                                                                       |   |       |
|-------|-----|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---|-------|
|       |     |                              | 0 0 1 0: Water cooled                                                                                                                 |   |       |
| BYTE6 | 7-4 | Battery type                 | Bit7 Bit6 Bit5 Bit4<br>0 0 0 0: lithium iron phosphate<br>0 0 0 1: Ternary<br>0 0 1 0: Lithium manganate<br>0 0 1 1: Lithium titanate |   |       |
| BYTE7 | 7-0 | Total available power (kWh). | 1                                                                                                                                     | 0 | 0-255 |

#### 5.2.5 BMS sends instrument messages 5

| ID          | P   | PGN                                          |            |            |                     | His | Period (ms). |
|-------------|-----|----------------------------------------------|------------|------------|---------------------|-----|--------------|
| 0x 1805A6F0 | 6   | R                                            | DP         | PF         | PS                  | 240 | 500ms        |
|             |     | 0                                            | 0          | 05         | 16<br>6             |     |              |
| Data        |     |                                              |            |            |                     |     |              |
| location    |     | The data name                                | resolution | The offset | The range of values |     |              |
| BYTE        | BIT |                                              |            |            |                     |     |              |
| BYTE0       | 7-0 | Rechargeable energy storage subsystem number | 1          | 0          | 0-255               |     |              |
| BYTE1       | 7-0 | The current frame sequence number n          | 1          | 0          | 0-255               |     |              |
| BYTE2       | 7-0 | Section 3*n+0 battery voltage low bytes      | EV0.001    | 0          | 0-6V                |     |              |
| BYTE3       | 7-0 | Section 3*n+0 battery voltage high bytes     |            |            |                     |     |              |
| BYTE4       | 7-0 | Section 3*n+1 battery voltage low bytes      | EV0.001    | 0          | 0-6V                |     |              |
| BYTE5       | 7-0 | Section 3*n+1 battery voltage high bytes     |            |            |                     |     |              |
| BYTE6       | 7-0 | Section 3*n+2 battery voltage low bytes      | EV0.001    | 0          | 0-6V                |     |              |
| BYTE7       | 7-0 | Section 3*n+2 battery voltage high bytes     |            |            |                     |     |              |

#### 5.2.6 BMS sends instrument messages6

| ID          | P | PGN           |    |          |            | His                 | Period (ms). |
|-------------|---|---------------|----|----------|------------|---------------------|--------------|
| 0x 1806A6F0 | 6 | R             | DP | PF       | PS         | 240                 | 500ms        |
|             |   | 0             | 0  | 06       | 166        |                     |              |
| Data        |   |               |    |          |            |                     |              |
| location    |   | The data name |    | resoluti | The offset | The range of values |              |

| BYTE  | BIT |                                              | on  |     |           |
|-------|-----|----------------------------------------------|-----|-----|-----------|
| BYTE0 | 7-0 | Rechargeable energy storage subsystem number | 1   | 0   | 0-255     |
| BYTE1 | 7-0 | The current frame sequence number            | 1   | 0   | 0-255     |
| BYTE2 | 7-0 | 6*n+0 battery temperature                    | 1°C | -40 | -40-150°C |
| BYTE3 | 7-0 | 6*n+1 battery temperature                    | 1°C | -40 | -40-150°C |
| BYTE4 | 7-0 | 6*n+2nd battery temperature                  | 1°C | -40 | -40-150°C |
| BYTE5 | 7-0 | 6*n+3 battery temperature                    | 1°C | -40 | -40-150°C |
| BYTE6 | 7-0 | 6*n+4 battery temperature                    | 1°C | -40 | -40-150°C |
| BYTE7 | 7-0 | 6*n+5 battery temperature                    | 1°C | -40 | -40-150°C |

#### 5.2.7 BMS sends instrument messages 7

| ID          | P   | PGN                          |            |            |                                                                                                                  | His | Period (ms). |
|-------------|-----|------------------------------|------------|------------|------------------------------------------------------------------------------------------------------------------|-----|--------------|
| 0x 1807A6F0 | 6   | R                            | DP         | PF         | PS                                                                                                               | 240 | 500ms        |
|             |     | 0                            | 0          | 07         | 166                                                                                                              |     |              |
| Data        |     |                              |            |            |                                                                                                                  |     |              |
| location    |     | The data name                | resolution | The offset | The range of values                                                                                              |     |              |
| BYTE        | BIT |                              |            |            |                                                                                                                  |     |              |
| BYTE0       | 6-0 | Manufacturer number          | 1          | 0          | 0-99                                                                                                             |     |              |
| BYTE0       | 7   | Battery part number code     | 1          | 0          | 0-1<br>[0 is without heating, 1 is with heating].                                                                |     |              |
| BYTE1       | 7-0 | Date of manufacture (year).  | 1          | 0          | 0: 2000<br>1: 2001<br>2: 2002<br>And so on                                                                       |     |              |
| BYTE2       | 7-0 | Date of manufacture (month). | 1          | 0          | 0: Invalid<br>1: 1 month<br>2: February<br>3: March<br>By analogy, the maximum is 12, greater than 12 is invalid |     |              |
| BYTE3       | 7-0 | Date of manufacture (day).   | 1          | 0          | 0: Invalid<br>1:1 day<br>2:2 days<br>3:3 days<br>By analogy, the maximum is 31, greater than 31 is invalid       |     |              |

|       |     |                                    |   |   |        |
|-------|-----|------------------------------------|---|---|--------|
| BYTE4 | 7-0 | Production serial number low bytes | 1 | 0 | 0-9999 |
| BYTE5 | 7-0 | Production serial number high byte |   |   |        |
| BYTE6 | 7-0 | The encoding length                | 1 | 0 | 0-99   |
| BYTE7 | 7-0 | retain                             | 1 | 0 | Set FF |

### 5.2.8 BMS on-board charger message 1

| ID          | P   | PGN                                                           |                                                                                      |            |                     | His | Period (ms). |
|-------------|-----|---------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|---------------------|-----|--------------|
| 0x 18FFE5F4 | 6   | R                                                             | DP                                                                                   | PF         | PS                  | 240 | 1000ms       |
|             |     | 0                                                             | 0                                                                                    | 255        | 229                 |     |              |
| Data        |     |                                                               |                                                                                      |            |                     |     |              |
| location    |     | The data name                                                 | resolution                                                                           | The offset | The range of values |     |              |
| BYTE        | BIT |                                                               |                                                                                      |            |                     |     |              |
| BYTE0       | 7-0 | The maximum allowable voltage at the charging terminal is low | 0.1V                                                                                 | 0          | 0-500V              |     |              |
| BYTE1       | 7-0 | The maximum allowable charge-side voltage is high             |                                                                                      |            |                     |     |              |
| BYTE2       | 7-0 | The maximum allowable charge current is low bytes             | 0.1A                                                                                 | 0          | 0-600A              |     |              |
| BYTE3       | 7-0 | The maximum allowable charge current is high                  |                                                                                      |            |                     |     |              |
| BYTE4       | 0   | control                                                       | Bit0<br>0: The charger starts charging;<br>1: Battery protection, turn off charging. |            |                     |     |              |
| BYTE4       | 7-1 | retain                                                        | 1                                                                                    | 0          | All set 1           |     |              |
| BYTE5       | 0   | Charging completion status                                    | Bit0<br>0: Not completed;<br>1: Charging is complete.                                |            |                     |     |              |
| BYTE5       | 7-1 | retain                                                        | 1                                                                                    | 0          | All set 1           |     |              |
| BYTE6       | 7-0 | retain                                                        | 1                                                                                    | 0          | Set FF              |     |              |
| BYTE7       | 7-0 | retain                                                        | 1                                                                                    | 0          | Set FF              |     |              |

### 5.3.1 On-board chargers send broadcast messages 1

| ID          | P | PGN |    |     |    | His | Period (ms). |
|-------------|---|-----|----|-----|----|-----|--------------|
| 0x 18FF50E5 | 6 | R   | DP | PF  | PS | 229 | 1000ms       |
|             |   | 0   | 0  | 255 | 80 |     |              |
| Data        |   |     |    |     |    |     |              |

| location |     | The data name                     | resolution                                                                                                                                  | The offset | The range of values |
|----------|-----|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------|
| BYTE     | BIT |                                   |                                                                                                                                             |            |                     |
| BYTE0    | 7-0 | The output voltage is low bytes   | 0.1V                                                                                                                                        | 0          | 0-500V              |
| BYTE1    | 7-0 | The output voltage is high bytes  |                                                                                                                                             |            |                     |
| BYTE2    | 7-0 | The output current is low bytes   | 0.1A                                                                                                                                        | 0          | 0-600A              |
| BYTE3    | 7-0 | The output current is high bytes  |                                                                                                                                             |            |                     |
| BYTE4    | 0   | Hardware failure                  | Bit0<br>0: Normal;<br>1: Fault.                                                                                                             |            |                     |
| BYTE4    | 1   | The charger temperature is faulty | Bit0<br>0: Normal;<br>1: High temperature protection.                                                                                       |            |                     |
| BYTE4    | 2   | Low voltage limited power mode    | Bit0<br>0: The input voltage is normal;<br>1: The input voltage is low and enters the low power mode.                                       |            |                     |
| BYTE4    | 3   | Input voltage status              | Bit0<br>0: The input voltage is normal;<br>1: Input voltage overvoltage or undervoltage fault.                                              |            |                     |
| BYTE4    | 4   | Output overcurrent                | Bit0<br>0: Normal;<br>1: Output overcurrent.                                                                                                |            |                     |
| BYTE4    | 5   | Startup status                    | Bit0<br>0: Charger off state;<br>1: Charging status.                                                                                        |            |                     |
| BYTE4    | 6   | Communication status              | Bit0<br>0: Normal;<br>1: Receive timed out.                                                                                                 |            |                     |
| BYTE4    | 7   | Battery connection status         | Bit0<br>0: The battery connection is normal;<br>1: The battery is reversed or not connected.                                                |            |                     |
| BYTE5    | 7-0 | Charger temperature               | 1℃                                                                                                                                          | -40        | -40-215℃            |
| BYTE6    | 7-0 | retain                            | 1                                                                                                                                           | 0          | Set FF              |
| BYTE7    | 3-0 | retain                            | 1                                                                                                                                           | 0          | 1 in full           |
| BYTE7    | 6-4 | Charging phase                    | Bit6 Bit5 Bit4<br>0 0 0: Constant current charging<br>0 0 1: Constant voltage charging<br>0 1 0: Constant power charging<br>Other undefined |            |                     |
| BYTE7    | 7   | Charging status                   | Bit7<br>0: Not charged;<br>1: Charging.                                                                                                     |            |                     |

Note: When the BMS detects the CC signal or receives the vehicle charger packet 0x18FF50E5, it will send the message at 1S interval 0x18FFE5F4 to the vehicle charger. After receiving the information, the charger works according to the voltage and current setting of the message data. If the two parties do not receive each other's packets within 5S, the communication timeout fault is entered and the output is turned off.

#### 5.4.1 Instrument Broadcast Message1

| ID          | P   | PGN                              |    |            |            | His                 | Period (ms). |
|-------------|-----|----------------------------------|----|------------|------------|---------------------|--------------|
| 0x 180150A6 | 6   | R                                | DP | PF         | PS         | 166                 | 100ms        |
|             |     | 0                                | 0  | 01         | 80         |                     |              |
| Data        |     |                                  |    |            |            |                     |              |
| location    |     | The data name                    |    | resolution | The offset | The range of values |              |
| BYTE        | BIT |                                  |    |            |            |                     |              |
| BYTE0       | 7-0 | Total mileage low bytes          |    | 0.1m       | 0          | 0- 429496729. 5m    |              |
| BYTE1       | 7-0 | Total mileage is low high bytes  |    |            |            |                     |              |
| BYTE2       | 7-0 | Total mileage high and low bytes |    |            |            |                     |              |
| BYTE3       | 7-0 | Total mileage is high in bytes   |    |            |            |                     |              |
| BYTE4       | 7-0 | retain                           |    | 1          | 0          | Set FF              |              |
| BYTE5       | 7-0 | retain                           |    | 1          | 0          | Set FF              |              |
| BYTE6       | 7-0 | retain                           |    | 1          | 0          | Set FF              |              |
| BYTE7       | 7-0 | retain                           |    | 1          | 0          | Set FF              |              |